

Maya Breanne Flannery

Updated: 2025-06-06

PhD Candidate | [Department of Psychology, Neuroscience & Behaviour](#) | McMaster University

Current research position at the [BEATLab](#)

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1. Education

Doctor of Philosophy (in progress)

2022–Present

· McMaster University, Dept. of Psychology, Neuroscience & Behaviour, ON, Canada. · Areas of study: Psychology, music cognition and perception: Development of multi-person physiological measures and analyses of cardiac activity.

Master of Science

2020–2022

· McMaster University, Dept. of Psychology, Neuroscience & Behaviour, ON, Canada. · Thesis: *From algorithm to cognition: Music acoustic features and their perceptual correlates*

Honours Bachelor of Arts

2016–2020

· McMaster University, Dept. of Psychology, Neuroscience & Behaviour, ON, Canada. · Major: Music Cognition · Minor: Music

2. Research Career

Researcher | Brain/body Attention Entrainment & Timing (BEAT) Lab

03/2023–Present

· Formal modelling of music preference with respect to personality, mood, familiarity, complexity, and music acoustic features. Development of open-source multi-person cardiac sensors, metrics, and analyses for use in naturalistic environments. · Supervised by Dr. Lauren Fink.

Researcher | Digital Music Lab

09/2019–03/2023

· Developed and tested manipulable and objectively measurable *music acoustic features* for use in music classification and description. Investigated correlation of personality factors and music preference with systematic manipulation of musical stimuli. · Supervised by Dr. Matthew Woolhouse.

Research Assistant | Digital Music Lab

01/2018–09/2019

· Examined music listening behaviour with analysis of a large dataset. Investigated memory for musical key; assisted with creation of musical stimuli; conducted experiments with participants; programmed analyses of data. · Supervised by Dr. Joanna Spyra and Dr. Matthew Woolhouse.

Research Assistant | Large Interactive Virtual Environment (LIVE) Lab

05/2016–09/2017

· Supported research projects with design, set up, execution, and clean up of experiments. Demonstrated lab technology: Active acoustics, Yamaha Disklavier piano, motion capture, live physiological responses. Led tours for general public, colleges, high schools, day camps. Supported live performances: 10dB Concert series. · Supervised by Dr. Laurel Trainor

2.1 Primary Research Skills

Experiment Design

· Experienced with design and implementation of single-participant in-person and online lab studies consisting of listening tasks and self-report and physiological measures (obtaining ethics approval, study promotion, survey/questionnaire design, MIDI and audio music manipulation, UI development).

· Experienced with design and implementation of multi-person (40+ participants) live concert studies; involving the continuous collection of self-report and physiological (eye-tracking and cardiac) responses from audience members.

Programming and Data Analysis

· Languages: Python (pandas, plotly, matplotlib, scikit-learn, optuna, kivy, PsychoPy), R (tidyverse, shiny, rstan), TypeScript/JavaScript (JsPsych, node.js).

· Analysis techniques: Linear/logistic regression, t-tests, ANOVAs (mixed, repeated measures); natural language processing (NLP); feature selection (PCA), nearest neighbors, decision trees, gradient boosting; Bayesian inference.

· Software: Linux/Mac/Windows environments, Git, Docker, LaTeX, Markdown, Bitwig Studio, Logic pro, Reason, MuseScore.

Science Communication

· Focused on open science principles—producing accessible and easy-to-understand research studies that are well-documented, reproducible, and ethically designed.

· Shared research through posters, presentations, and publications in departmental, community, and field-specific academic conferences.

3. Publications

Manuscripts in preparation

Flannery, M.B., & Fink, L. (*In prep*). Naturalistic measurement of multi-person cardiac activity using open source smartwatch technology.

Flannery, M.B., Schlichting, J., Saxena, S., & Fink, L. (*In prep*). Investigating the influence of contextual information on cardiac activity in response to musical performance.

Publications

Flannery, M. B., & Woolhouse, M. H. (2021). Musical preference: Role of personality and music-related acoustic features. *Music & Science*, 4, 20592043211014014.

Bansal, J., **Flannery, M. B., & Woolhouse, M. H.** (2021). Influence of personality on music-genre exclusivity. *Psychology of Music*, 49(5), 1356-1371.

Software

Flannery, M. B., & Fink, L. (2025). BEATmonitor (Version v0.2.250427) [Computer software]. <https://github.com/beatlab-mcmaster/BEATmonitor>

4. Conference Talks & Presentations

Conference Presentations

Flannery, M. B., & Fink, L. (2025, July 21–25). *Multi-person photoplethysmography using open-source smartwatch technology Development, validation, and applications in naturalistic music listening*. International Conference on Music Perception and Cognition 18. [Virtual] São Paulo, Brazil.

Flannery, M. B., & Fink, L. (November, 2024). *Naturalistic measurement of multi-person cardiac activity using open source smartwatch technology* [Poster]. NeuroMusic 20. Hamilton, ON, Canada.

Flannery, M. B., Schlichting, J., Saxena, S., & Fink, L. (July, 2024). *Investigating the influence of contextual information on cardiac activity in response to musical performance* [Talk]. Society for Music Perception and Cognition, Banff, AB, Canada.

Saxena, S., **Flannery, M. B.**, Schlichting, J., & Fink, L. (July, 2024). *Gaze behavior in online and in-person concert and film viewing: A large-scale naturalistic eye-tracking study* [Talk]. Society for Music Perception and Cognition, Banff, AB, Canada.

Schlichting, J., **Flannery, M. B.**, Saxena, S., & Fink, L. (July, 2024). *Social justice advocacy through music performance: Do contextual information and social co-presence shape audience reactions and behavior?* [Poster]. Society for Music Perception and Cognition, Banff, AB, Canada.

Flannery, M. B., Schlichting, J., Saxena, S., & Fink, L. (April, 2024). *Investigating the influence of contextual information on cardiac activity in response to musical performance* [Poster]. Graduate Research Day 2024. Hamilton, ON, Canada.

Flannery, M. B., & Fink, L. (October 2023). *Automating music stimuli creation and analyses: A music synthesis algorithm for producing ground truth data* [Poster]. NeuroMusic 19. Hamilton, ON, Canada.

Flannery, M.B., Fink, L., & Woolhouse, M. H. (August, 2023). *Models trained on procedurally generated stimuli predict human judgments of Music Acoustic Features in real-world music*. The 17th International Conference on Music Perception and Cognition. [Virtual] Tokyo, Japan.

Flannery, M.B., & Woolhouse, M. H. (November, 2022). *Music acoustic features: Do machine predictions correspond to human judgments?* [Poster]. The 18th Annual McMaster NeuroMusic Conference. Hamilton, Canada.

Flannery, M. B., Anant, S., & Woolhouse, M. H. (November, 2021). *A musical snapshot: What does popular music sound like?* The 17th Annual McMaster NeuroMusic Conference. [Virtual] Hamilton, Canada.

Flannery, M. B., & Woolhouse, M. H. (July, 2021). *Music-related acoustic features predict music preference*. The 16th International Conference on Music Perception and Cognition. [Virtual] UK/Australia/India/Mexico.

Flannery, M. B., & Woolhouse, M. H. (February, 2021). *Why do I like this song? Music acoustic features and beyond*. Graduate Research Day, [Virtual] Hamilton, Canada.

Flannery, M. B., & Woolhouse, M. H. (November, 2020). *Introducing MAFs: Music acoustic features and their relationship to personality*. The 16th Annual McMaster NeuroMusic Conference. [Virtual] Hamilton, Canada.

Guest Lectures

Phillips, E., Pazdera, J., & **Flannery, M. B.** (2022, 2023, 2024). *The replication crisis and the open science movement*. Presented to PNB 2XT0 - Psychology, Neuroscience & Behaviour Tutorial. Hamilton, Canada.

Flannery, M. B. (November, 2023). *Modelling the music listening experience: How do we measure the listener and the music?* PNB Coggie Talk. Hamilton, Canada.

Flannery, M. B. (September, 2023). *Open Science Fundamentals*. Presented to PNB 3EE3 - Perception Lab. Hamilton, Canada.

Flannery, M. B., & Hashemi, A. (October, 2022). *Introduction to the Analysis of Variance (ANOVA)*. Presented to PNB 3XE3 - Inferential Statistics and Research Methods. Hamilton, Canada.

Phillips, E., Pazdera, J., & **Flannery, M. B.** (December, 2021). *The open science movement*. Presented to PNB 2XT0 - Psychology, Neuroscience & Behaviour Tutorial. Hamilton, Canada.

Flannery, M. B. (November, 2021). *Data formatting and visualization*. Presented to PNB 2A03 - Python for Psychology, Neuroscience & Behaviour. Hamilton, Canada.

Workshops

Flannery, M. B., Saxena, S., Schlichting, J., & Fink, L. (November 2024). *Cardiac signals*. Signals, Sensors, and Sounds Workshop. Hamilton, Canada.

Flannery, M. B. (December, 2023). *Considerations for Selecting MIR Tools*. McMaster Music Information Retrieval (MIR) Workshop. Hamilton, Canada.

Flannery, M. B. (December, 2023). *Predicting Musical Cues*. McMaster Music Information Retrieval (MIR) Workshop. Hamilton, Canada.

Flannery, M. B. (April, 2023). *Cleaning up messy data with Python and pandas*. Data Analysis Support Hub (DASH) Workshop. Hamilton, Canada.

Community Outreach

Swierczek, K., Spyra, J., & **Flannery, M. B.** (2022). *A tale of two tones: Musical dissonance in perspective*. Science Rendezvous. Hamilton, Canada.

5. Teaching Career

Teaching assistantships

01/2025

· PNB 2XE3: Descriptive Statistics and Research Methods

09/2022 & 09/2024

· PNB 3XE3: Inferential Statistics and Research Methods

09/2023 & 01/2024

· PNB 3EE3: Perception Lab

05/2023

· HUMBEHV 3MD3: Research Methods for Applied Psychology in Human Behaviour

01/2023

· PSYCH 2MP3: Introduction to Music Cognition

01/2022

· PNB 2XF3: Perspectives in Psychology, Neuroscience & Behaviour

09/2021

· PNB 2A03: Python for Psychology, Neuroscience & Behaviour

01/2021

· PNB 2XD3: Integrative Psychology, Neuroscience & Behaviour Through Scientific Writing

09/2020

· PSYCH 2B03: Personality Psychology

Mentoring & Supervision

Advised experiment design and analyses for undergraduate independent projects and theses:

2024–2025

· Thesis: *Maladaptive music listening: A systematic review and meta-analysis* · Thesis: *Differential Effects of Active and Passive Bilateral Stimulation [Eye Movement Desensitization and Reprocessing; EMDR] on Distress Levels*

2023–2024

· Thesis: *Semantic analyses of music descriptors: An investigation of music tags and descriptors in a large dataset*

2022–2023

· Thesis: *Music preference and skip-behaviour: The relationship between listening time and confidence in music preference ratings*

2021–2022

· Thesis: *Discrimination of music features: Determining the saliency of music acoustic features*

6. Previous Work Experience

Consultant | Data Analysis Support Hub (McMaster University)

2022

· Supported McMaster University students and faculty with statistical analyses in R, Python, and Excel. Developed and led workshops on data on reproducible analyses.

7. Professional Development

Certifications

2025

· Introduction: Advanced Research Computing · Scaling HPC Workflows · Network Analysis of Neurophysiological Data

2022

· Teaching and Learning: Theory and Inquiry and Essential skills and Digital Literacies

2020

· Data Science Workshop Series & Certification

Journal Clubs

2020–Present

· ReproducibiliTea Journal Club · Music Cognition and Auditory Perception Journal Club

Committees

2023

· Graduate Research Day Committee

2017

· 2SLGBTQIA+ Peer Support Volunteer

8. Grants & Scholarships

· SSHRC Canada Graduate Scholarship - Master's (2021)

· Graduate Studies Scholarship (2020)

· University Senate Scholarship (2019)

9. References

- Please contact me to request references.